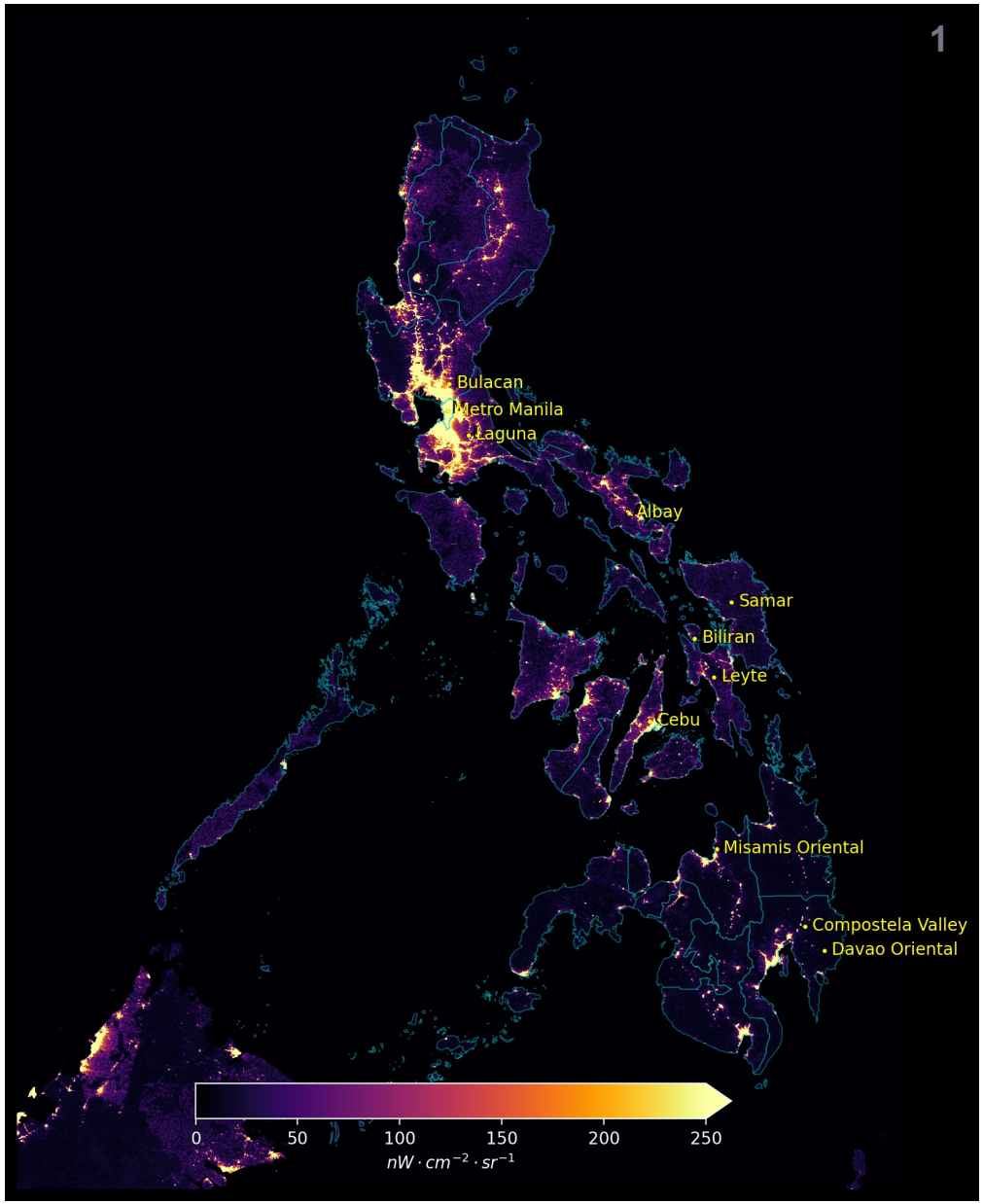
# PhD Research Meeting 45

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# 1. Monthly Meeting 3 Synthesis

| Issue | What remains unresolved |
|---|---|
| Pre-event cyclicity | Its period, magnitude, spatial consistency, and relationship to observation conditions or actual light emission |
| Baseline | Whether the 60-day median, a longer static reference, or a dynamic reference supports defensible recovery estimates |
| Electricity evidence | Whether local historical records exist, who holds them, and what can be released |
| Physical EO | Available dates, locations, products, access conditions, and the recovery dimension they measure |
| Recovery families | Whether membership remains stable under baseline and observability changes |
| RQ2 scope | The primary comparison endpoint, intended audience, and role of survey or demographic evidence |

# 2. Institutional routes to granular electricity load data

| Route | Connection to the data request |
|---|---|
| **Philippine Space Agency (PhilSA):** Atty. Paul Balite | Endorsement linking the inquiry to ERC and NEA contacts |
| **Energy Regulator Commission (ERC):** Engr. Alvin Jones Ortega | Regulatory route to distribution utilities and their data custodians |
| **National Electrification Administration (NEA):** Engr. Eric Villar | Route through electric-cooperative oversight and coordination |
| **Philippine Rural Electric Cooperatives Association, Inc. (PHILRECA):** Ms. Ella | Complementary route through its network of member cooperatives |
| **Wholesale Electricity Spot Market (WESM):** Drazel Reyes | Market Studies contact for inquiries about demand records and relevant data holders |

**Progress:** Direct cooperative emails/phone numbers and data requests have expanded into institutional referral routes, including Atty. Balite's endorsement.

**Immediate action:** Request granular electricity load data through these contacts and proceed based on their responses.

# Reviewer 1: Methodological robustness and clarity

- **30Dcentred smoothing:** justify temporal scale; test shorter windows; clarify retros pective vs near-real-time interpretation
- **Composite score:** test sensitivity to the 0.6 / 0.1 / 0.3 weighting
- **Scientific contribution:** distinguish journal paper clearly from preliminary publicat ion
- **Temporal completeness:** explicitly state $\tau$ used in Table 2 and Figure 5
- **Figure 3:** improve NTL contrast stretch
- **Figure 6:** make colour direction intuitive and consistent
- **Figures 7–8:** reduce line thickness/overlap and improve readability

# Reviewer 2: Novelty, validation and scope

- **Novelty:** contribution beyond combining established completeness, stability and alignment metrics is unclear
- **Parameter robustness:** sensitivity analysis requested for composite score and methodological parameters
- **Transferability:** five Visayas sub-grids do not establish broader tropical generalisability
- **Event evidence:** requests validation across multiple independent disasters
- **Application performance:** asks whether qualification actually improves outage detection/recovery estimation
- **30-day smoothing:** may inflate long-term alignment while suppressing rapid disruptions



# Reviewer 3: Framing, parameter selection and uncertainty

- Strengthen **abstract**: technical gap, specific novelty and quantitative findings
- Present **numbered contributions** linked directly to the identified gaps
- Clarify **50% diagnostic threshold vs threshold sweep** and how preferred thresholds are selected
- Test **composite-score weight sensitivity**; discuss rather than necessarily implement adaptive thresholds
- Address implications of the **30-day window / temporal land-cover changes**
- Explain **high correlation despite only ~21–50% retained coverage**
- Add **statistical uncertainty** around correlation estimates
- Expand **limitations, failure cases, generalisability and future work**



# Editor decision: what actually needs to change

**Editor's three overarching concerns**

- Distinctiveness of the scientific contribution
- Robustness of key methodological choices
- Adequacy of validation relative to the claims

**Proposed revision strategy**

- Strengthen the **scientific increment** beyond the preliminary publication
- Run a bounded **robustness package** for smoothing, weights, and thresholds
- Add **uncertainty + retained-data context**
- Narrow claims where validation does not support broader generalisation
- Fix definitions, figures, and manuscript framing
- **Do not automatically turn RQ1 into a new outage-detection or multi-event study**



# Scientific contribution and novelty

**POSITION: AGREE — this needs substantially better framing**

- The individual metrics (SC, TC, dropout, CV, correlation) are **not themselves novel**
- Contribution should instead be the **diagnostic-first admissibility framework**
- Observability is assessed **before** downstream interpretation, rather than treated as preprocessing metadata

Integrates:

- direct vs gap-filled observations, pixel- and settlement-scale diagnostics, settlement-stratified reliability, explicit completeness–coverage trade-offs, systematic configuration selection, regional replication across five sub-grids
- Explicitly separate what was already shown in the **preliminary publication** from what the journal paper newly establishes
- New robustness analyses should become part of the journal paper's **incremental scientific contribution**



# Composite score and threshold selection

**POSITION: AGREE on sensitivity; CLARIFY what the score means**

- Current weights 0.6*r + 0.1*RMSE + 0.3*N are explicitly heuristic
- Run a **small, declared weight-sensitivity analysis**
- Ask whether conclusions are stable at two levels: **settlement-mask choice,** exact **completeness threshold**

Important distinction:

  - **50%** = majority-support reference used for some observability diagnostics
  - **10–90% sweep** = load-alignment sensitivity experiment
- Avoid language such as **“optimal threshold”,** Prefer: **“highest-scoring configuration under stated weighting priorities”**
- If exact rankings are unstable but urban masks remain stable, **that itself is the result**. **Adaptive thresholding is not required for this revision**; it is a possible future extension

# High correlation vs limited temporal coverage

**POSITION: AGREE — this should become a central result, not a weakness we hide**

- Selected configurations retain only approximately **21–50% of dates**
- High *r* therefore means: strong agreement **when admissible observations exist**, **not** reliable daily/everyday monitoring
- Always report together: r, RMSE, retained *N* / coverage, dropout structure
- Show whether retained observations are temporally/seasonally clustered
- Add uncertainty around correlations with a method respecting time dependence
- **Block bootstrap** is more defensible than independent-day significance testing
- Clarify that Figure 6's **50% colour saturation is a visual reference, not a hard maximum**



# Do we need to demonstrate better outage detection?

**POSITION: DISAGREE with expanding the study; CLARIFY the scope**

- The manuscript explicitly states: **“The objective is not event detection per se.”**
- RQ1 asks when reliability-qualified NTL tracks **independent electricity-load trends**
- The analysis follows the **full 2012–2024 record**, rather than selecting only disaster windows, and Haiyan is primarily a **stress-test / interpretive example**
- Rai is also visible in the long-term record, but neither event constitutes a formal detector benchmark
- Therefore: no need to build a new outage detector, no need to prove improved recovery estimation against competing methods
- Instead, revise language that could be read as claiming those outcomes
- Event detection and recovery metrics remain a **downstream application once reliability conditions are established**



# Do we need multiple disasters/other tropical countries‘

**POSITION: PARTLY AGREE — narrow the claim, not necessarily expand the dataset**

- Five Visayas sub-grids demonstrate **regional replication**
- They do **not** demonstrate universal tropical transferability
- Strongest transferable finding appears to be: urban-focused settlement masks repeatedly perform well, precise completeness threshold remains context-dependent
- Therefore distinguish: **transferable framework / selection logic**, **locally validated parameter values**
- Reframe “transferability” as **conditional portability / regional consistency** where appropriate
- Global completeness analysis supports the **observability problem**, not global load-validation performance
- No new countries are necessary if claims are narrowed accordingly

# Land-cover change and the GHSL epoch

**POSITION: NEEDS CLARIFICATION — reviewer combines two different issues**

- **Temporal smoothing issue:** a long moving window can mix changes occurring within that period
- **Settlement-mask issue:** a static GHSL epoch is applied across the longer 2012–2024 record
- These are related but **not the same bias**
- Shorter-window sensitivity addresses temporal mixing
- Static GHSL should be acknowledged as a separate **epoch-mismatch limitation**
- GHSL is used here for **structural stratification**, not to infer land-cover change
- A full dynamic land-cover reconstruction is **not necessary** unless the paper claims to attribute changes to evolving settlement morphology



# Straightforward revisions + proposed revision boundar

## Straightforward changes

- Improve Figure 3 radiance contrast
- Explicitly document $\tau$ for Table 2 and Figure 5
- Standardise Figure 6 colour direction
- Simplify/thin Figures 7–8
- Rewrite abstract with quantitative results **plus coverage/temporal-scale caveats**
- Add numbered contributions
- Strengthen limitations and explicit failure cases

## What I propose we actually add

- Temporal-window sensitivity
- Composite-weight / threshold sensitivity
- Correlation uncertainty + retained-date diagnostics
- Explicit contribution comparison with preliminary publication

## What I do *not* propose adding unless we retain broader claims

- New outage-detection algorithm
- Multi-disaster performance benchmark
- New-country validation
- Adaptive threshold algorithm
- Dynamic land-cover reconstruction

